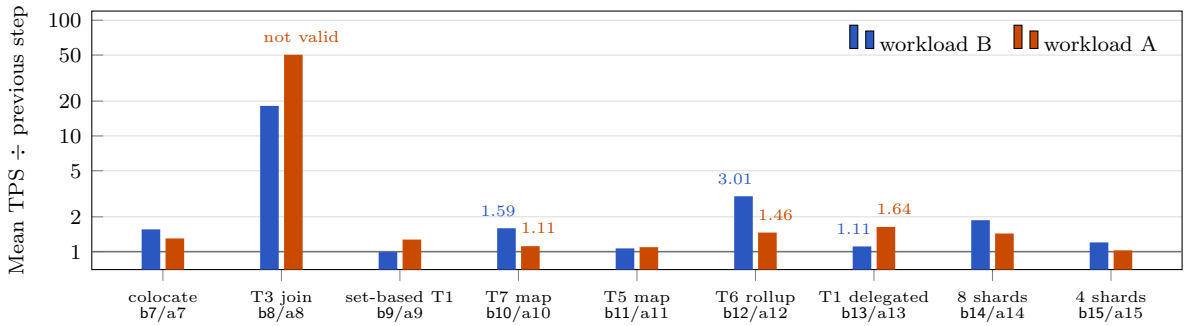


Figure 1: The same nine steps on both workloads: mean TPS of each step divided by the step before. T7 routing and the rollup pay more on B, where T7 is 30% and T6 10% of the mix (5% each on A); delegating T1 pays more on A, where T1 is 36% of the mix (10% on B). The A runs used a different node set, so only the ratios are compared; a8 is not valid from 4 clients up (most clients aborted).

	TPS							Mean of 3	
Clients	b19	b20	b21	Mean	SE	95% CI	CV (%)	p99 (ms)	Retried
1	279.5	277.0	276.0	**277.5**	1.1	273.0–282.0	0.66	14.2 ± 0.04	0
4	1,309.5	1,302.2	1,287.1	**1,299.6**	6.6	1,271.1–1,328.1	0.88	13.7 ± 0.06	185
8	2,803.7	2,810.7	2,806.2	**2,806.9**	2.0	2,798.1–2,815.6	0.12	12.7 ± 0.03	949
16	3,573.1	3,578.6	3,573.8	**3,575.2**	1.7	3,567.8–3,582.6	0.08	19.4 ± 0.03	2,138
32	3,958.7	3,965.5	3,986.4	**3,970.2**	8.3	3,934.3–4,006.1	0.36	34.6 ± 0.17	4,231
64	4,020.7	4,036.7	4,060.1	**4,039.2**	11.4	3,989.9–4,088.4	0.49	73.7 ± 0.29	7,532
Mean	2,657.5	2,661.8	2,664.9	**2,661.4**	2.1	2,652.2–2,670.6	0.14		

Table 1: The final design, run three times (b19 on 2026-09-28, b20 and b21 on 2026-10-11). SE is the standard error of the mean of the three runs; the 95% interval uses Student's t with 2 degrees of freedom. p99 is the 99th-percentile latency and Retried the number of transactions pgbench retried at least once, both as mean ± SE or mean of the three runs.

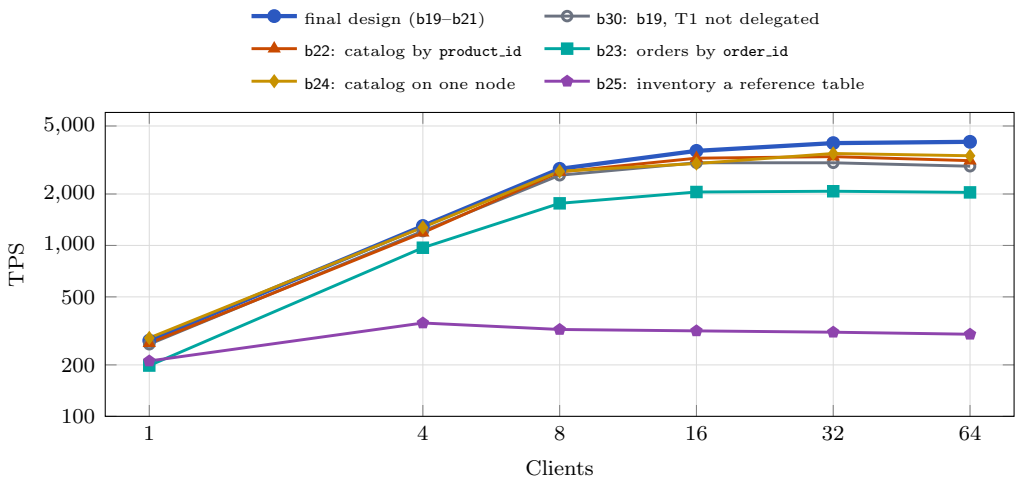


Figure 2: The four alternative placements against the final design (error bars: 95% interval of the three runs, smaller than the markers). b22, b24 and b25 move `inventory` off the merchant's worker, so their T1 cannot be delegated; b30, the ablation cell with only T1 delegation off, isolates that change. b22 and b24 beat b30 but not the final design; b25 serialises every `inventory` write.

mean TPS		T6 rollup / T7 routing			
T5	T1	on/on	off/on	on/off	off/off
on	on	100	17	20	10
on	off	82†	18†	20	11
off	on	85	17	21	11
off	off	75†	10†	21	9†

TPS at 64 clients		T6 rollup / T7 routing			
T5	T1	on/on	off/on	on/off	off/off
on	on	100	15	14	8
on	off	72†	16†	15	8
off	on	73	13	15	8
off	off	65†	8†	17	8†

Table 2: The 2^4 ablation around the final design: TPS as a percentage of the final design's (mean of b19–b21), averaged over the six client counts (left) and at 64 clients (right). Rows switch T5 routing and T1 delegation, columns the T6 rollup and T7 routing; the top-left cell is the final design. Darker is faster (logarithmic shading). † marks cells with aborted clients: T1's commit failed on a `SERIALIZABLE` conflict, which pgbench does not retry.

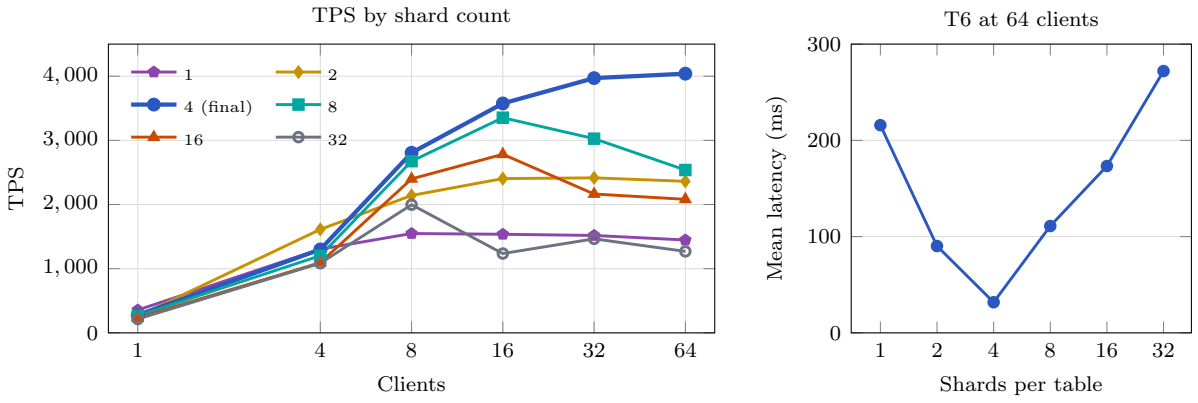


Figure 3: The final design with 1 to 32 shards per table (4 is the final design; error bars: 95% interval of its three runs). Left: fewer shards win at low load, because fewer transactions commit across workers, but put all the data on one or two workers. Right: T6 runs one task per shard and is cheapest with one shard per worker.